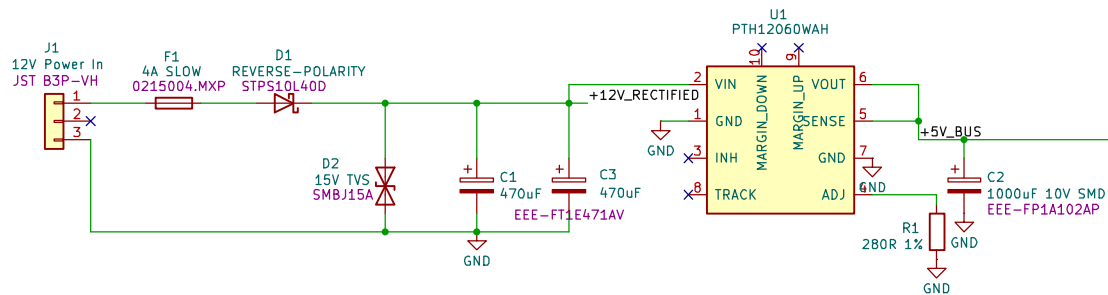


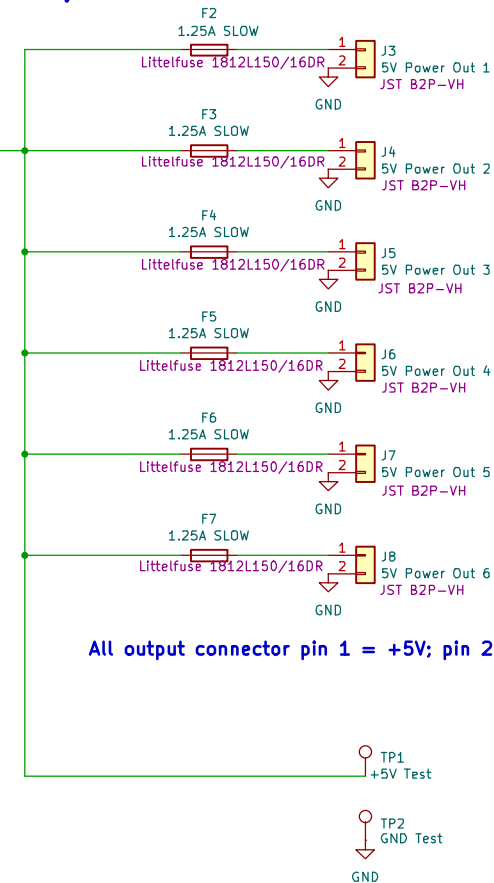
12 V INPUT / PROTECTION

TI PTH12060WAH BUCK MODULE (10 A)

SIX FUSED 5 V OUTPUTS



U1: TI PTH12060WAH through-hole buck module.
R1 = 280 ohm sets 5.0 V. Leave INHIBIT, TRACK,
MARGIN UP and MARGIN DOWN open for normal operation.



All output connector pin 1 = +5V; pin 2 = GND.

DESIGN NOTES:

1. Board distributes power for one panel column (six branches of 56 LEDs).
2. U1 is TI PTH12060WAH, configured for regulated 5.0 V output.
3. F1 and F2-F7 are starting values only; confirm from measured operating/inrush current.
4. Use 2 oz copper, broad +5V/GND pours, short high-current paths, and 2.0 mm FR-4.
5. D1 must be selected/rated for full input current; an ideal-diode MOSFET stage may replace it.
6. Verify every connector polarity against actual harness before assembly.
7. Board outline is 300 mm x 100 mm; confirm hinge and cable clearance before fabrication.